

The respiratory tract includes the nose, sinuses, pharynx, larynx, trachea, lungs, and alveoli. Gas exchange occurs within the lungs at the alveolar membrane. The composition of the inhaled atmospheric air is approximately 80% nitrogen, 20% oxygen, and only 0.05% carbon dioxide. However, due to gas exchange the composition of exhaled air is approximately 80% nitrogen, 15% oxygen, and 5% carbon dioxide.

The total volume of the lungs is separated into four distinct fractions: residual volume (RV), tidal volume (TV), expiratory reserve volume (ERV), and inspiratory reserve volume (IRV). *TV* is the volume of air exchanged in the lungs in a single breath during normal breathing. *RV* is the volume not exchanged because the lungs are never fully emptied of air. *ERV* is the volume that can be emptied during maximal exhalation. Similarly, *IRV* is the volume that can be filled during maximal inhalation.

These lung volumes can be combined to describe other functional parameters. For example, the vital capacity *VC* is given by:

$$VC = IRV + TV + ERV$$

Equation 1

In addition, the functional reserve capacity *FRC* is given by:

$$FRC = RV + ERV$$

Equation 2

Researchers measured these lung parameters in healthy subjects, and the results are summarized in Table 1.

Typically, the air inhaled into the lungs is warmed as it is in close contact with the surfaces of the respiratory tract. Researchers measured a heat capacity of 0.40 J/K for the volume of air exchanged in the lungs during normal breathing.

Table 1 Lung Parameters Measured in Healthy Individuals

Parameter	Volume (mL)
Residual volume	1,500
Expiratory reserve volume	1,000
Tidal volume	500
Inspiratory reserve volume	2,500
Vital capacity	4,000
Functional reserve capacity	2,500

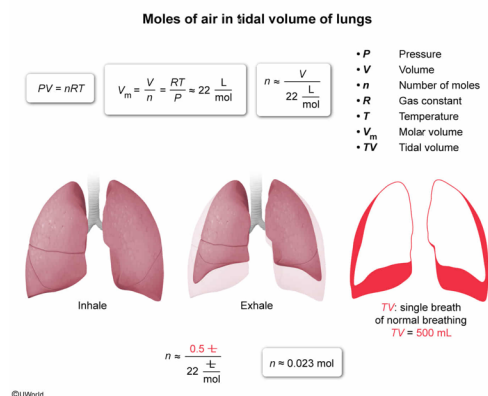
What is the number of moles of gas exchanged in a single breath during normal breathing? (Note: Assume all gases are ideal and at STP.)

- ✓ ☒ A. 0.023 mol [48%]
☐ B. 0.045 mol [24%]
☐ C. 0.068 mol [15%]
☐ D. 0.11 mol [12%]

Correct

47% answered correctly

Explanation:



The **ideal gas law** states that the product of the pressure P and volume V of an ideal gas is equal to the product of the number of moles n , the **gas constant** R , and the **temperature** T :

$$PV = nRT$$

Standard temperature and pressure (STP) is defined as $T = 273 \text{ K}$ and $P = 1 \text{ atm}$. At STP, the ratio of V and n is a constant called the **molar volume** V_m , given by:

$$V_m = \frac{V}{n} = \frac{RT}{P} = \frac{(0.082 \frac{\text{L} \cdot \text{atm}}{\text{K} \cdot \text{mol}})(273 \text{ K})}{1 \text{ atm}} \approx 22 \frac{\text{L}}{\text{mol}}$$

This equation can be rearranged to solve for n :

$$n = \frac{V}{V_m} \approx \frac{V}{22} \text{ mol} \approx 0.045 \cdot V \text{ mol}$$

In this question, the volume of gas exchanged in a single breath during normal breathing is the **tidal volume** TV , given as 500 mL in [Table 1](#). Converting the units of TV to liters yields:

$$V = \frac{500 \text{ mL}}{1,000 \frac{\text{mL}}{\text{L}}} = 0.5 \text{ L}$$

Therefore, n is approximately equal to:

$$n \approx (0.045)(0.5) \approx 0.023 \text{ mol}$$

(Choice B) 0.045 mol is obtained by using the expiratory reserve volume in the equation, not the tidal volume.

(Choice C) 0.068 mol results from using the residual volume of the lungs, not the tidal volume.

(Choice D) 0.11 mol is calculated by using the inspiratory reserve volume instead of the tidal volume.

Educational objective:

The molar volume of an ideal gas at standard temperature and pressure is equal to 22 L/mol.

Time spent: 0 sec

Subject:
Physics

QID: 404117

System:
Fluids

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Functional reserve capacity	2,500

Which of the following expressions is equal to the residual volume *RV* of the lungs?

- ☐ A. $FRC - IRV - TV$ [5%]
☐ B. $FRC + IRV + TV$ [1%]
☐ C. $FRC - IRV - TV - VC$ [19%]
☒ D. $FRC + IRV + TV - VC$ [73%]

Correct

73% answered correctly

Explanation:

Explanation.

Solving equations for reserve volume

$$FRC = RV + ERV$$
$$VC = IRV + TV + ERV$$

- **FRC** Functional reserve capacity
- **RV** Reserve volume
- **ERV** Expiratory reserve volume
- **VC** Vital capacity
- **IRV** Inspiratory reserve volume
- **TV** Tidal volume

Solve for **RV** and **ERV**: $RV = FRC - ERV$
 $ERV = VC - IRV - TV$

Combine equations: $RV = FRC - (VC - IRV - TV)$

$$RV = FRC + IRV + TV - VC$$

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In this question, **RV** is calculated from **Equation 2** as:

$$RV = FRC - ERV$$

Furthermore, **Equation 1** can be **solved** for **ERV**, giving:

$$ERV = VC - IRV - TV$$

Substituting the equation above into the equation for **RV** yields:

$$RV = FRC - (VC - IRV - TV)$$

$$RV = FRC - VC + IRV + TV$$

$$RV = FRC + IRV + TV - VC$$

Therefore, **RV** is equal to $FRC + IRV + TV - VC$.

(Choices A and B) These equations omit the vital capacity term from Equation 1.

(Choice C) This equation subtracts the inspiratory reserve volume and tidal volume terms instead of adding them.

Educational objective:

The equations for vital capacity and functional reserve capacity can be combined and rearranged to solve for any of the variables.

Time spent: 0 sec

Subject:
Physics

QID: 404118

System:
Fluids

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Typically, the air inhaled into the lungs is warmed as it is in close contact with the surfaces of the respiratory tract. Researchers measured a heat capacity of 0.40 J/K for the volume of air exchanged in the lungs during normal breathing.

Table 1 Lung Parameters Measured in Healthy Individuals

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Vital capacity	4,000
Functional reserve capacity	2,500

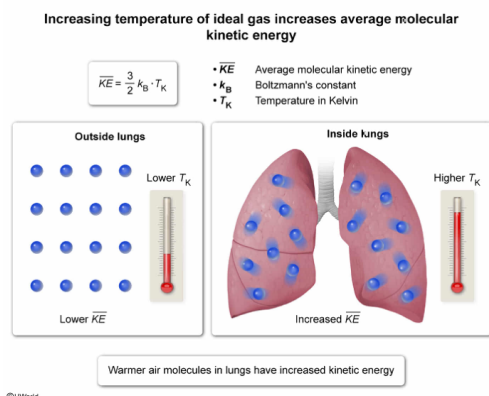
That the temperature of the inspired air increases inside the lungs means that the air molecules: (Note: Assume that air behaves like an ideal gas.)

- ☐ A. are closer together. [3%]
- ☐ B. experience increased intermolecular forces. [4%]
- ✓ ☒ C. have increased average kinetic energy. [90%]
- ☐ D. have decreased average velocity. [1%]

Correct

91% answered correctly

Explanation:



Gas molecules are uniformly distributed within the volume of the gas and experience a uniform pressure. However, the molecules themselves are in continuous random motion. The **kinetic molecular theory of gases** states that the average molecular **kinetic energy** $\overline{KE}$ of **ideal gas** molecules is directly proportional to the product of **Boltzmann's constant** k_B and the **temperature** T_K in Kelvin:

$$\overline{KE} = \frac{3}{2} k_B \cdot T_K$$

In this question, air taken in during respiration is warmed within the lungs. Therefore, the increased temperature of the air leads to a proportionate increase in the average kinetic energy of the gas particles.

(Choice A) Air molecules are closer together when the volume decreases. However, temperature is directly related to the average molecular kinetic energy, not the volume.

(Choice B) Intermolecular forces are assumed to be negligible in an ideal gas, and independent of temperature.

(Choice D) A decrease in the velocity of the air molecules decreases the average kinetic energy, which corresponds to a decrease in temperature, not an increase in temperature.

Educational objective:

According to the kinetic molecular theory of gases, increasing the temperature of an ideal gas increases the average kinetic energy of the gas molecules.

Time spent: 0 sec

Subject:
Physics

QID: 404119

System:
Fluids

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Parameter	Volume (mL)
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Tidal volume	500
Inspiratory reserve volume	2,500
Vital capacity	4,000
Functional reserve capacity	2,500

During normal breathing, the temperature of the air inhaled into the lungs in a single breath increases by 10 °C. How much heat is transferred into the volume of air in a single breath?

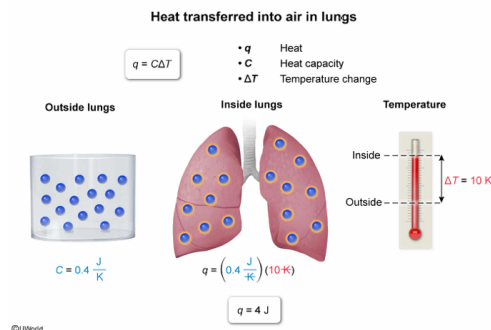
- ☐ A. 0.40 J [8%]
- ✓ ☒ B. 4.0 J [70%]
- ☐ C. 5.0 J [13%]
- ☐ D. 25 J [7%]

Correct

70% answered correctly

Explanation:

Explanation.



The **calorimetry equation** states that the **heat** q transferred into an object equals the product of the object's mass m , the specific heat c , and the change in **temperature** ΔT :

$$q = mc \Delta T$$

The value of c corresponds to the thermal properties of a unit mass of the material. An object with an arbitrary m can be described by its particular **heat capacity** C , which equals the product of the object's m and the value of c for the material comprising the object:

$$C = mc$$

Combining these two equations gives:

$$q = C \Delta T$$

In this question, ΔT of the air is 10°C . Because the **temperature units** $^\circ\text{C}$ and K differ only by a constant offset, a change of 10°C is equivalent to a change of 10 K. Furthermore, the passage states that C for a volume of air inhaled during normal breathing equals 0.40 J/K . Substituting these values into the above equation yields:

$$q = \left(0.40 \frac{\text{J}}{\text{K}}\right)(10 \text{ K}) = 4.0 \text{ J}$$

Therefore, 4.0 J of heat is transferred into the volume of air inhaled during normal breathing.

(Choice A) 0.40 J is the heat required to increase the temperature of the air by 1 K, not 10 K.

(Choice C) 5.0 J results from incorrectly calculating the product of the air volume and the change in temperature.

(Choice D) 25 J is obtained by incorrectly calculating the ratio of the heat capacity and the change in temperature.

Educational objective:

Heat capacity is the amount of heat required to change the temperature of a particular object by 1 K. The heat transferred equals the product of the heat capacity of the object and the change in temperature.

Time spent: 0 sec
Subject:
Physics

QID: 404120
System:
Fluids

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Residual volume	1,500
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Vital capacity	4,000
Functional reserve capacity	2,500

The partial pressure of O₂ in expired air is: (Note: Assume all gases are ideal and at STP.)

- ☐ A. 608 mmHg. [6%]
- ☐ B. 152 mmHg. [27%]
- ☒ C. 114 mmHg. [57%]
- ☐ D. 38.0 mmHg. [8%]

Correct

58% answered correctly

Explanation:

Partial pressure of oxygen in expired air

$$P_g = X_g P_M$$
$$P_g = V_g P_M$$

- P_g Partial pressure of O_2
- X_g Mole fraction of O_2
- P_M Pressure of gas mixture
- V_g Percent volume of O_2



$P_M = 760 \text{ mmHg}$

$X_g = V_g = 15\%$

$P_g = 0.15(760 \text{ mmHg})$

$P_g = 114 \text{ mmHg}$

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According to [Dalton's law of partial pressures](#), the **partial pressure** P_g of a gas within a mixture of gases is equal to the pressure the gas would exert if it were alone in the same volume as the mixture. Furthermore, P_g equals the product of the [mole fraction](#) X_g of the gas and the [total pressure](#) P_M of the gas mixture:

$$P_g = X_g P_M$$

X_g is the ratio of the number of moles of the gas to the total number of moles in the mixture. For an [ideal gas](#), X_g equals the amount of the gas as a percentage of the **total volume**.

In this question, the passage states that O_2 is 15% of expired air by volume, and therefore X_g equals 0.15. Furthermore, P_M equals 760 mmHg because the gas mixture is at STP. Hence, P_g is given by:

$$P_g = (0.15)(760 \text{ mmHg}) = 114 \text{ mmHg}$$

(Choice A) 608 mmHg is obtained by using the percentage of N_2 in expired air instead of O_2 .

(Choice B) 152 mmHg is the partial pressure of O_2 in inspired air, not expired air.

(Choice D) 38.0 mmHg is the partial pressure of CO_2 in expired air, not O_2 .

Educational objective:

The partial pressure of a gas in a mixture of several gases is equal to the pressure it would exert if it were alone and occupying the same volume as the mixture. The gas's partial pressure depends on its mole fraction, which for ideal gases equals its volume fraction.

Time spent: 0 sec
Subject:
Physics

QID: 404121
System:
Fluids

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Functional reserve capacity	2,500

A way to increase the transfer of heat from the body to the environment would be to:

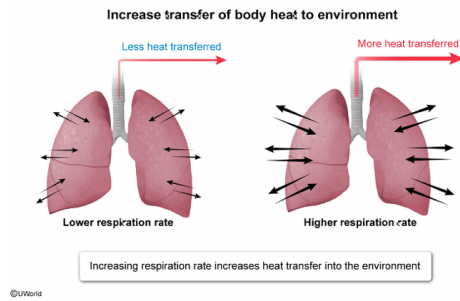
- ✔ ☒ A. increase the respiration rate. [82%]
☐ B. increase the residual volume of the lungs. [7%]
☐ C. decrease the alveolar surface area. [6%]
☐ D. decrease the blood flow in pulmonary vessels. [3%]

Correct

82% answered correctly

Explanation:

Explanation.



Body heat can be transferred into the environment through [conduction](#), [convection](#), or [radiation](#).

Conduction is achieved through physical contact whereas **convection** relies on the movement of a medium (eg, blood, air) to move the heat. Respiration and blood flow play significant roles in [transferring heat](#) from the body into the environment.

[Ventilation](#) transfers heat to the environment through conduction as the [pulmonary blood vessels](#) transfer heat into the air in the lungs, followed by convection as the air is exhaled. The conduction of heat can be increased by increasing the respiration volume or [dilating](#) pulmonary blood vessels. The convection of heat can be increased by increasing the respiration volume or rate.

Therefore, increasing the **respiration rate** increases the transfer of heat to the environment.

(Choice B) Increasing the residual volume of the lungs increases the lung volume that is not ventilated, which would not increase heat transfer to the environment.

(Choices C and D) Decreasing the alveolar surface area or the pulmonary blood flow would reduce the conduction of heat from the pulmonary vessels to the air in the lungs, decreasing heat transfer to the environment.

Educational objective:

The transfer of body heat to the environment is increased by increasing the respiration rate, respiration volume, blood flow in pulmonary vessels, or a combination of these parameters.

Time spent:0 sec

Subject:
Physics

QID:404122

System:
Fluids